

SankeyLogic for Excel: Technical & Security Compliance Dossier

Version: 4.1

Developer: Visual Analytics Ltd.

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Reference: ISO/IEC 27001 A.12.1.2 & NIST SP 800-53 compliant structure.

1. The Engine Behind Decision Intelligence

Security, Speed, and Deterministic Precision

SankeyLogic is a **native analytical engine** that operates directly within the Microsoft Excel process. Unlike web-based solutions, SankeyLogic executes as native machine code, bypassing the rendering limitations and security overhead of browser-based environments.

- **Standalone and Native:** The core propagation logic is built on a memory-safe, compiled architecture that functions without external dependencies such as WebView2 or Edge.
- **Zero Latency:** All calculations are performed locally in the workstation's RAM. There are no cloud-based wait times or API bottlenecks.
- **Weighted Propagation:** The system simulates the flow of impacts and risks through the network, supporting model depths of up to 15 levels.
- **Deterministic Precision (SSOT):** The mathematical layer is strictly decoupled from the visual layer. Every Bézier curve on the canvas is mathematically guaranteed to match the mass-balance and concentration risk (HHI) calculated in the core.

Volatile In-Memory Execution (Data Sovereignty)

Unlike traditional add-ins that store sensitive analytical metadata or large configuration blobs within the Excel .xlsx file (increasing file corruption risks and leaving permanent data footprints), SankeyLogic operates on a **Volatile In-Memory** principle.

All analytical graphs and mathematical models exist only within the active session's protected memory space. Upon closing the workbook, the analytical state is purged.

Upon re-opening a workbook, an explicit model re-generation is required, guaranteeing that no 'stale' or out-of-sync visuals can influence decision-making. It guarantees:

1. **Zero Data Leakage:** No proprietary logic or sensitive model parameters are ever written to the disk or stored inside the file.
2. **File Integrity:** Your workbooks remain 100% standard Excel files, free from third-party binary bloat or XML corruption.
3. **Live Accuracy:** Every time the model is activated, it is re-validated against the current source data, preventing the use of "stale" or out-of-sync visualizations.

2. Compliance Overview: Executive Snapshot

The design of SankeyLogic deliberately avoids external services, background processes, and runtime dependencies to simplify enterprise evaluation.

Domain	Compliance Statement
Execution Model	100% local; no cloud or network dependencies
Security	No external communication, telemetry, or persistent state
Privacy	No personal data processed or transmitted (GDPR out of scope)
Integrity	Build integrity verified via SHA-256 checksum
Governance	GPO-deployable; versioned and auditable build manifests
Licensing	Standalone tool; no recurring subscription-check calls

3. System Architecture and IT Governance

3.1 Local Execution & Security Boundaries

- **Process Isolation:** The add-in operates strictly within the EXCEL.EXE process space.
- **Network Air-Gap:** No internet, intranet, or external API connections are ever established, eliminating network exposure risks.
- **Zero Telemetry:** The add-in performs no remote logging or background activity, preventing unintended data transmission.
- **Uninstallation:** Removal is equivalent to file deletion - no residual files, configuration data, or registry entries remain on the system.

3.2 Deployment (IT Friendly)

- **No Admin Rights:** The add-in requires no administrative privileges, supporting deployment in restricted or managed enterprise environments.
- **RegFree Architecture:** It does not modify the Windows Registry or alter system-wide configurations.
- **GPO Support:** Installation is supported in both local and read-only network directories governed by Group Policy.

4. Security Framework Alignment (ISO & NIST)

The following table maps SankeyLogic controls to international security standards:

Framework	Control Reference	Implementation in SankeyLogic
ISO 27001	A.12.1.2 Change Management	Controlled release process with versioned build manifests.
ISO 27001	A.9.2.3 User Responsibilities	Operates under standard user privileges; no elevation required.

Framework	Control Reference	Implementation in SankeyLogic
NIST SP 800-53	SI-7 Software Integrity	Build integrity verified via SHA-256 checksum.
NIST SP 800-53	PL-8 Information Security Arch.	Fully local execution with no external data flows.
GDPR	Art. 25 Privacy by Design	No personal data is collected, transmitted, or processed.

5. Software Integrity and Verification

Each release is produced from human-readable source code under controlled versioning. Verification is handled through a deterministic model:

1. **Integrity Verification:** Integrity is verified via a published SHA-256 checksum.
2. **Auditability:** Source code access may be provided for security review under controlled conditions.
3. **Deterministic Output:** The build procedure ensures that the binary output is verifiable and tamper-evident.

6. Data Protection and GDPR Considerations

The add-in performs no processing, storage, or transmission of personal data as defined in GDPR Article 4(1).

- **No Data Processor Role:** Visual Analytics Ltd. acts neither as Data Controller nor as Data Processor under GDPR.
- **Data Minimization:** Product design inherently complies with GDPR principles through its offline-only architecture.
- **No DPIA Required:** A Data Protection Impact Assessment is typically not required due to the absence of personal-data processing.

Appendix: Integrity Verification (SHA-256)

Enterprise IT can independently verify the authenticity of the SankeyLogic.dll using standard PowerShell tools without relying on external certificate authorities:

```
PowerShell
Get-FileHash .\SankeyLogic.dll -Algorithm SHA256
```

Contact & Legal Information Visual Analytics Ltd. | 7 Bell Yard, London, WC2A 2JR | contact@chartengine.io

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